



Chapter 1: Introduction to Derivatives Markets (Continued)

6. Types of Traders (Section 1.6)

- **Market Success Factor:** The outstanding success of derivatives markets is attributed to high **liquidity**, which results from attracting diverse types of traders.
- **Three Broad Categories:**
 - i. **Hedgers:** Use derivatives to **reduce/mitigate risk**.
 - ii. **Speculators:** Use derivatives to **bet on the future direction** of a market variable (taking a position).
 - iii. **Arbitrageurs:** Use derivatives to **lock in a riskless profit** by taking offsetting positions in multiple markets.
- **Hedge Funds (Business Snapshot 1.3):**
 - **Definition:** Similar to mutual funds, but accept funds only from **financially sophisticated individuals** and do not publicly offer securities.
 - **Regulation:** They are **relatively free of regulations** applied to mutual funds (like limits on leverage or mandatory immediate redemption). This allows them to develop sophisticated, proprietary strategies.
 - **Fees:** Typically high (e.g., 1–2% of AUM plus 20% of profits).
 - **Strategies:** Use derivatives extensively for all three purposes (hedging, speculation, and arbitrage). Examples include:
 - **Long/Short Equities:** Small exposure to overall market direction.
 - **Convertible Arbitrage:** Long convertible bond + actively managed short underlying equity.
 - **Global Macro:** Trades based on anticipated global macroeconomic trends.

7. Hedgers (Section 1.7)

a. Hedging Using Forward Contracts

- **Goal:** Forward contracts are used to **neutralize risk by fixing the price** of the underlying asset to be paid or received in the future.
- **ImportCo Example (Long Hedge):** A US company owes £10 million in 3 months.
 - **Action:** The company enters a **3-month forward contract to buy GBP** at the quoted offer price (\$1.5538/GBP).
 - **Result:** It fixes the USD cost at \$15,538,000, eliminating the risk of the GBP strengthening (exchange rate rising).
- **ExportCo Example (Short Hedge):** A US company will receive £30 million in 3 months.
 - **Action:** The company enters a **3-month forward contract to sell GBP** at the quoted bid price (\$1.5533/GBP).
 - **Result:** It fixes the USD receipt at \$46,599,000, eliminating the risk of the GBP weakening (exchange rate falling).
- **Key Aspect:** Hedging **reduces risk**, but there is **no guarantee** that the outcome with hedging will be better than the outcome without it.

b. Hedging Using Options

- **Goal:** Options provide **insurance**. They protect against adverse price movements while allowing the hedger to **benefit from favorable price movements**.
- **Difference from Forwards:** Options involve the payment of an **up-front fee** (the premium), whereas forwards generally do not.
- **Investor Example (Portfolio Protection):** An investor owns 1,000 shares at \$28/share but fears a decline.
 - **Action:** Buy 10 July **put option contracts** ($K = \$27.50$, cost \$1 per share). Total cost is \$1,000.
 - **Result:** The investor guarantees the ability to sell the shares for at least \$27.50 during the option's life.
 - If the price falls to \$20, the investor exercises the option, selling at \$27.50. Net amount realized (after premium): \$26,500.
 - If the price rises to \$35, the options expire worthless, but the stock holding is worth \$35,000 (a gain, minus the \$1,000 premium cost).

8. Speculators (Section 1.8)

- **Goal:** Speculators intentionally seek **exposure to adverse/favorable movements** in an asset's price to profit from their market view.

a. Speculation Using Futures

- **Mechanism:** Futures provide **leverage**, allowing a speculator to take a large position with a relatively **small initial outlay** (called **margin**).
- **Sterling Example (Long Speculation):** A speculator believes the GBP will strengthen.
 - **Spot Alternative:** Buy £250,000 spot (\$386,750 investment).
 - **Futures Alternative:** Buy 4 April futures contracts (£62,500 each). Initial investment is only the margin (e.g., \$20,000).
- **Comparison:** Both result in large profits if the exchange rate rises, but the futures trade requires much less capital up front, magnifying the return on the investment (leverage).

b. Speculation Using Options

- **Mechanism:** Options also provide **leverage**, magnifying financial consequences for a given investment.
- **Stock Example (Call Option Speculation):** Stock is \$20. Speculator invests \$2,000.
 - **Buy Stock:** Purchase 100 shares. If the price rises to \$27, the profit is **\$700**.
 - **Buy Options:** Purchase 2,000 call options ($K = \$22.50$, premium \$1). If the price rises to \$27, the payoff is $(\$27 - \$22.50) \times 2,000 = \$9,000$. Net profit: **\$7,000**. (10 times more profitable).
- **Risk Profile:**
 - **Gain:** Good outcomes become very good.
 - **Loss:** The speculator's loss is **limited to the amount paid for the options** (the premium, \$2,000 in this case).
- **Futures vs. Options for Speculation:**
 - **Futures:** Potential loss and potential gain are both **very large**.
 - **Options:** Potential loss is **limited** to the initial premium paid.

9. Arbitrageurs (Section 1.9)

- **Definition:** Arbitrage involves **locking in a riskless profit** by simultaneously entering into offsetting transactions in two or more markets.
- **No Arbitrage Assumption:** The existence of arbitrageurs ensures that in practice, **only very small disparities** in prices can exist. Most theoretical arguments regarding derivatives pricing assume that **no arbitrage opportunities exist**.
- **Stock Exchange Example:**
 - Stock price is \$150 in New York (NYSE).
 - Stock price is £100 in London (LSE).
 - Exchange rate is \$1.5300/£.
 - **Arbitrage Action:** Simultaneously buy 100 shares in New York (cost \$15,000) and sell 100 shares in London (receive £10,000).
 - **Profit:** Converting the £10,000 to USD yields \$15,300.
 - **Risk-Free Profit:** $\$15,300 - \$15,000 = \$300$ (ignoring transaction costs).
- **Market Impact:** The actions of arbitrageurs (buying low, selling high) quickly drive the prices back into equilibrium, making the opportunity disappear almost instantly.